

Q1. What are the different data sources we can connect to Power BI? Submit Screenshot for any 3 data connectivity.

Power BI can connect to a wide variety of data sources. These sources can be broadly divided into the following categories:

A. File Sources

- Excel Workbook
- Text/CSV
- XML
- JSON
- PDF
- Folder
- Parquet
- SharePoint Folder

B. Database Sources

- SQL Server
- Microsoft Access
- Oracle
- MySQL
- PostgreSQL
- IBM Db2
- SAP HANA
- Snowflake
- Amazon Redshift
- Google BigQuery
- Teradata

C. Online Services

- SharePoint Online

- Microsoft Dynamics 365
- Salesforce
- Azure services
- Google Analytics
- Microsoft Exchange
- Other online services

D. Other Sources

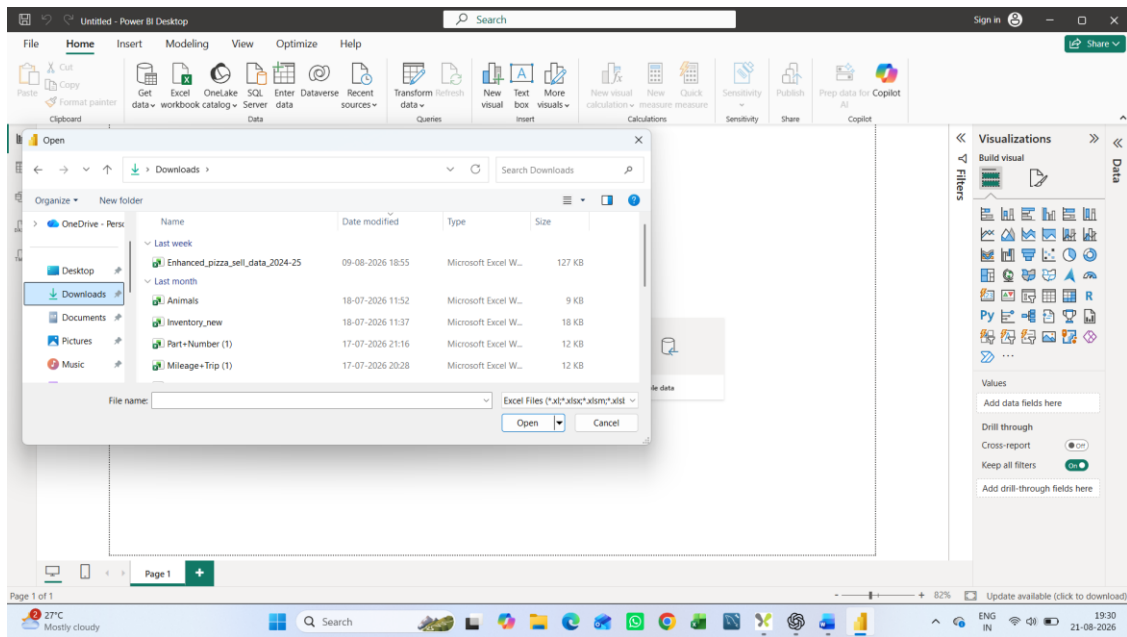
- Web
- OData Feed
- ODBC
- OLE DB
- Azure Data Lake
- Power Platform sources

Three Data Connectivity Screenshots

Screenshot 1 – Excel Workbook

Go to:

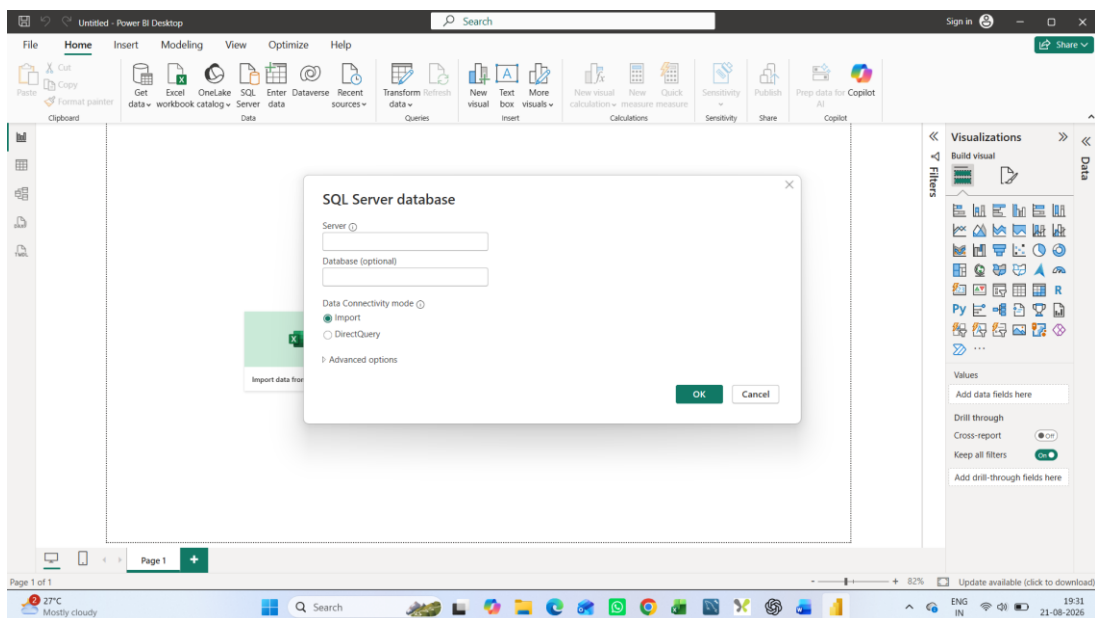
Home → Get Data → Excel Workbook



Screenshot 2 – SQL Server

Go to:

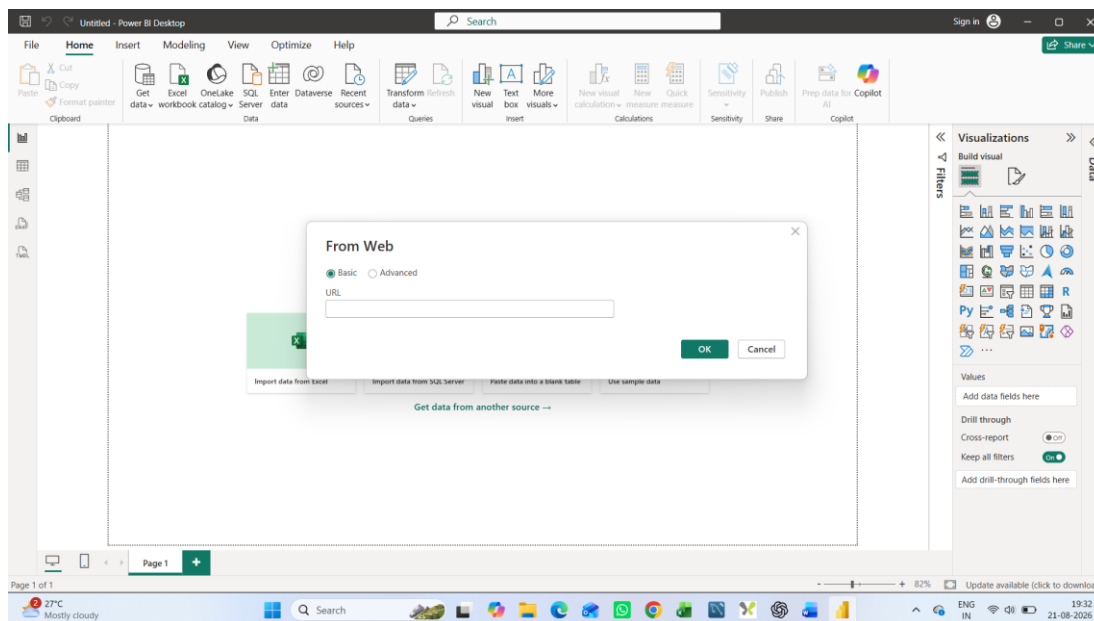
Home → Get Data → SQL Server



Screenshot 3 – Web

Go to:

Home → Get Data → Web



Q2. What is the new name of Power BI dataset?

The new name of **Power BI Dataset** is "**Semantic Model**."

Microsoft renamed the Power BI dataset content type to **Semantic Model**. The change was made because "semantic model" better represents the data model used by Power BI for reporting and analytics.

Q3. What are the different components of Power BI Desktop Tool?

The main components of Power BI Desktop are:

1. Report View

Report View is used to create interactive reports and dashboards. We can add charts, graphs, tables, cards, slicers, and other visualizations here.

2. Table/Data View

Table View is used to see the data that has already been loaded into the Power BI model. It is useful for inspecting rows and columns and creating measures and calculated columns.

3. Model View

Model View displays tables and the relationships between them. It is used to create, edit, and manage relationships between tables.

4. Power Query Editor

Power Query Editor is used for connecting to data sources and cleaning and transforming data before loading it into the Power BI model.

5. DAX

DAX stands for **Data Analysis Expressions**. It is used to create measures, calculated columns, and other calculations for data analysis.

6. Visualizations Pane

The Visualizations area provides different types of charts and visuals such as bar charts, column charts, line charts, pie charts, maps, cards, and tables.

7. Data/Fields Pane

The Data or Fields pane displays the tables, columns, and measures available in the Power BI model. These fields can be dragged into visualizations.

Q4. Explain in brief the working of each component in Power BI.

1. Report View – Creating Reports

Report View is the main area where we create reports. We select fields and add visualizations such as charts, tables, cards, and slicers.

Working:

Data → Select Fields → Choose Visual → Create Report

2. Table/Data View – Viewing Data

Table View allows us to inspect the data after it has been loaded into the Power BI model. We can check rows, columns, data types, measures, and calculated columns.

Working:

Loaded Data → Table View → Inspect Data → Create/Manage Calculations

3. Model View – Managing Relationships

Model View shows the structure of the data model. It displays tables and their relationships. We can create or modify relationships between tables.

Working:

Tables → Identify Common Columns → Create Relationships → Build Data Model

Microsoft describes Model View as a graphical representation of tables, columns, and relationships in the model.

4. Power Query Editor – Cleaning and Transforming Data

Power Query Editor is used to prepare data before analysis. We can remove duplicates, remove blank rows, change data types, split columns, merge tables, filter data, and perform other transformations.

Working:

Connect to Data → Clean Data → Transform Data → Apply Changes → Load Data

Power Query Editor can connect to one or more data sources, shape and transform the data, and then load the resulting model into Power BI Desktop.

5. DAX – Performing Calculations

DAX is used to perform calculations and create measures and calculated columns.

For example:

Total Sales = SUM(Sales[Sales Amount])

This measure can then be used in charts, cards, and other visuals.

Working:

Data Model → Write DAX Formula → Create Measure/Column → Use in Report

6. Visualizations – Representing Data

Visualizations convert data into graphical formats. Different visuals can be selected depending on the type of analysis required.

For example:

- **Bar Chart** → Compare categories
- **Line Chart** → Show trends over time
- **Pie/Donut Chart** → Show proportions
- **Card** → Show a single important value
- **Map** → Show geographical information
- **Table/Matrix** → Display detailed information

7. Data/Fields Pane – Selecting Data

The Data/Fields pane contains tables, columns, and measures from the data model.

We can drag a field from this pane onto a visualization to display the required information.

Working:

Select Table → Select Field → Drag Field → Add to Visual → Analyze Data